

Interconnection Request: JAR Data Center at Palo Alto 110 kV station

Project description: 80 MW Data Center Facility

Developer: Google

Proposed Connection Point: Palo Alto 110 kV Substation

Grid Operator: California Independent System Operator (CAISO)

1. Project Overview and Scope

This document outlines the formal interconnection request for a new 80 MW load, identified as the JAR project. The project seeks to establish a new 110 kV connection to the CAISO-controlled grid at the Palo Alto 110 kV Substation.

The purpose of this document and the accompanying studies is to demonstrate the project's compliance with all applicable CAISO grid code requirements and reliability standards.

2. Summary of Analysis and Findings

Comprehensive interconnection studies (including Load Flow, Short Circuit, and Dynamic Stability analyses) have been completed to assess the impact of the 80 MW load on the transmission system.

The key findings are as follows:

- **Load Flow Study:** Under normal system conditions (N-0), the existing grid infrastructure is adequate to serve the 80 MW load without thermal violations on local transmission facilities.
- **Short Circuit Study:** The project's connection will not cause fault currents to exceed the interrupting ratings of existing circuit breakers at the Palo Alto 110 kV Substation. New, dedicated breakers for the project's connection will be installed as part of the facility build-out.
- **Stability & Voltage Study:** The stability analysis indicates that a load of this magnitude requires local reactive power support to maintain voltage stability and meet CAISO criteria during system contingencies (N-1 events). This analysis specifically modeled critical contingencies, including forced line outages and substation equipment failures at the Palo Alto 110 kV Substation.

3. Proposed Mitigation and Network Upgrades

To ensure compliance with CAISO reliability standards and mitigate the impacts identified in the stability analysis, the project will implement the following key mitigation measures and

network upgrades:

1. Installation of a 20 MVar STATCOM:

- **Purpose:** To provide dynamic voltage support.
- **Specification:** The STATCOM will be installed at the project's point of connection and will be configured to dynamically inject or absorb reactive power to maintain stable grid voltage within CAISO-required tolerances. This component is essential to prevent voltage instability during grid disturbances and is a core component of the project's design.

2. Facility-Side UPS Integration:

- **Purpose:** To ensure ride-through capability.
- **Specification:** The facility's internal Uninterruptible Power Supply (UPS) systems are designed to work in coordination with the grid. The STATCOM will further enhance the ability of the UPS systems to ride through grid events without interruption.

3. New 110 kV Connection:

- **Purpose:** To create the physical point of interconnection.
- **Specification:** The project will construct a new 110 kV transmission line from the Palo Alto 110 kV Substation to the data center, including all associated breakers, switches, and protection systems, built to CAISO and utility specifications.

4. Conclusion and Request for Interconnection

The completed studies demonstrate that the 80 MW data center project is feasible. The project sponsor commits to designing, procuring, and constructing all required mitigation measures, including the 20 MVar STATCOM.

With these measures in place, the facility will maintain a robust, reliable, and compliant connection to the CAISO grid. We formally request approval from CAISO to proceed with the interconnection as proposed.